

EDEXCEL INTERNATIONAL A LEVEL

WMA11/01 October 2021 Worked Solutions

Oct 2021 paper rebuilt with the worked solution after each question.

10

paper questions

WMA11

Pure 1

**Single-
paper
solutions**standalone IAL
route**Dr Eslam Ahmed**

Prepared for Dr Eslam Ahmed - eliteigcse.com

P1

PAST PAPER

WMA11/01 October 2021

Oct 2021 | 10 questions | 75 marks

10

questions

75

marks

Answers

worked solution
after each
question

Question 1

Integration

1. Find

$$\int 12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4} \, dx$$

giving each term in simplest form.

(5)

(Total 5 marks)

Worked Solution - Question 1

1. Rewrite with powers

$$12x^3 + \frac{1}{6\sqrt{x}} - \frac{3}{2x^4} = 12x^3 + \frac{1}{6}x^{-1/2} - \frac{3}{2}x^{-4}.$$

2. Integrate term by term

$$\int 12x^3 dx = 3x^4, \int \frac{1}{6}x^{-1/2} dx = \frac{1}{3}x^{1/2}, \text{ and } \int -\frac{3}{2}x^{-4} dx = \frac{1}{2}x^{-3}.$$

3. Write in simplest form

Therefore the integral is $3x^4 + \frac{\sqrt{x}}{3} + \frac{1}{2x^3} + c$.

Final answer

$$3x^4 + \frac{\sqrt{x}}{3} + \frac{1}{2x^3} + c.$$

Question 2**Differentiation**

2. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve has equation

$$y = 3x^5 + 4x^3 - x + 5$$

The points P and Q lie on the curve.

The gradient of the curve at both point P and point Q is 2

Find the x coordinates of P and Q .

(5)

(Total 5 marks)

Worked Solution - Question 2

1. Differentiate the curve

For $y = 3x^5 + 4x^3 - x + 5$, $\frac{dy}{dx} = 15x^4 + 12x^2 - 1$.

2. Set the gradient equal to 2

$15x^4 + 12x^2 - 1 = 2$, so $15x^4 + 12x^2 - 3 = 0$.

3. Reduce to a quadratic in x squared

Dividing by 3 gives $5x^4 + 4x^2 - 1 = 0$. Let $u = x^2$, giving $5u^2 + 4u - 1 = 0$.

4. Solve and reject the impossible value

$5u^2 + 4u - 1 = (5u - 1)(u + 1) = 0$. Since $u = x^2 \geq 0$, $u = \frac{1}{5}$.

5. Find x

$x^2 = \frac{1}{5}$, so $x = \pm \frac{1}{\sqrt{5}} = \pm \frac{\sqrt{5}}{5}$.

Final answer

$$x = \pm \frac{\sqrt{5}}{5}.$$

Question 3

Straight Line

3. (i) Solve

$$\frac{3}{x} > 4$$

(3)

(ii)

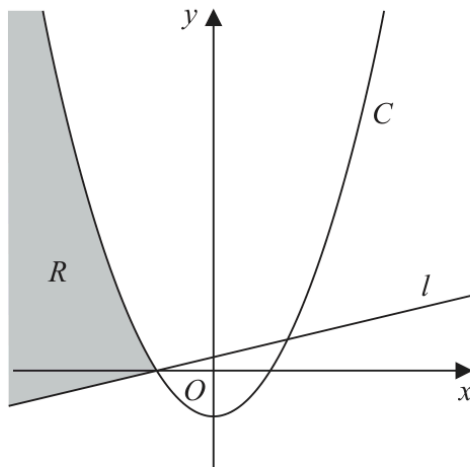


Figure 1

Figure 1 shows a sketch of the curve C and the straight line l .

The infinite region R , shown shaded in Figure 1, lies in quadrants 2 and 3 and is bounded by C and l only.

Given that

- l has a gradient of 3
- C has equation $y = 2x^2 - 50$
- C and l intersect on the negative x -axis

use inequalities to define the region R .

(3)

(Total 6 marks)

Worked Solution - Question 3

1. Solve the rational inequality

$\frac{3}{x} > 4$ is equivalent to $\frac{3 - 4x}{x} > 0$.

2. Use critical values

The critical values are $x = 0$ and $x = \frac{3}{4}$. A sign check gives the positive interval $x < 0$.

3. Find the equation of l

The curve C meets the negative x -axis when $2x^2 - 50 = 0$, so $x = -5$. Since l has gradient 3 and passes through $(-5, 0)$, $y = 3(x + 5) = 3x + 15$.

4. Define the shaded region

The shaded region lies above the line, below the curve, and to the left of their negative x -axis intersection. Hence $y > 3x + 15$, $y < 2x^2 - 50$, and $x < -5$.

Final answer

$\{(x, y) \mid y > 3x + 15, y < 2x^2 - 50, x < -5\}$.

Question 4

Trigonometric Functions

4.

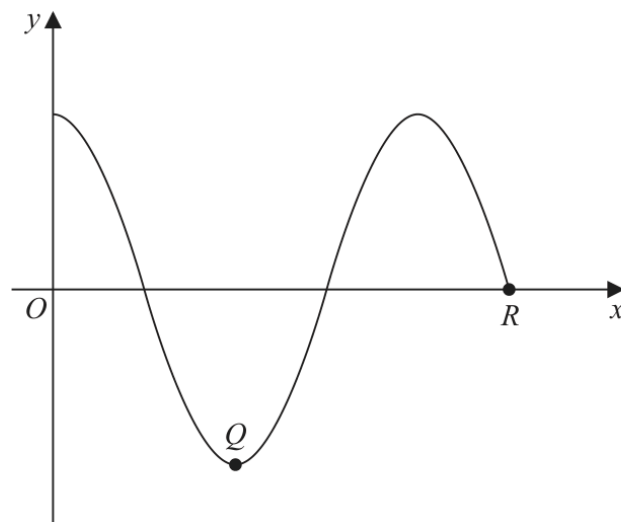


Figure 2

Figure 2 shows a sketch of the curve with equation $y = f(x)$, where

$$f(x) = \cos 2x^\circ \quad 0 \leq x \leq k$$

The point Q and the point $R(k, 0)$ lie on the curve and are shown in Figure 2.

(a) State

- (i) the coordinates of Q ,
- (ii) the value of k .

(3)

(b) Given that there are exactly two solutions to the equation

$$\cos 2x^\circ = p \quad \text{in the region } 0 \leq x \leq k$$

find the range of possible values for p .

(2)

(Total 5 marks)

Worked Solution - Question 4

1. Find Q

For $f(x) = \cos 2x^\circ$, the first minimum occurs when $2x = 180^\circ$, so $x = 90^\circ$ and $y = -1$. Therefore $Q = (90, -1)$.

2. Find k

The point $R(k, 0)$ is the x-intercept after the next maximum. This occurs when $2x = 450^\circ$, so $x = 225^\circ$. Hence $k = 225$.

3. Count horizontal intersections

A horizontal line $y = p$ gives exactly two intersections on $0 \leq x \leq 225$ when it cuts the two negative branches only, so $-1 < p < 0$.

4. Include the maximum case

There are also exactly two solutions when $p = 1$, at $x = 0$ and $x = 180$. Therefore $p = 1$.

Final answer

(a)(i) $Q = (90, -1)$. (a)(ii) $k = 225$.

(b) $-1 < p < 0$

Question 5

Straight Line

5. The line l_1 has equation $3y - 2x = 30$

The line l_2 passes through the point $A(24, 0)$ and is perpendicular to l_1

Lines l_1 and l_2 meet at the point P

- (a) Find, using algebra and showing your working, the coordinates of P .

(5)

Given that l_1 meets the x -axis at the point B ,

- (b) find the area of triangle BPA .

(3)

(Total 8 marks)

Worked Solution - Question 5

1. Rewrite l_1

$3y - 2x = 30$ gives $y = \frac{2}{3}x + 10$, so the gradient of l_1 is $\frac{2}{3}$.

2. Find l_2

Since l_2 is perpendicular to l_1 , its gradient is $-\frac{3}{2}$. Through $A(24, 0)$,
 $y = -\frac{3}{2}(x - 24) = -\frac{3}{2}x + 36$.

3. Solve for P

At the intersection, $\frac{2}{3}x + 10 = -\frac{3}{2}x + 36$. Multiplying by 6 gives
 $4x + 60 = -9x + 216$, so $13x = 156$ and $x = 12$.

4. Find the y-coordinate

$y = \frac{2}{3}(12) + 10 = 18$, so $P = (12, 18)$.

5. Find B

For the x-intercept of l_1 , set $y = 0$: $-2x = 30$, so $x = -15$. Hence $B = (-15, 0)$.

6. Find the triangle area

The base BA has length $24 - (-15) = 39$, and the height from P to the x-axis is 18. Area = $\frac{1}{2}(39)(18) = 351$.

Final answer

(a) $P = (12, 18)$.

(b) 351 square units.

Question 6

Graphs of Functions

6. In this question you must show all stages of your working.

Solutions relying on calculator technology are not acceptable.

A curve C has equation $y = f(x)$ where

$$f(x) = 2(x + 1)(x - 3)^2$$

(a) Sketch a graph of C .

Show on your graph the coordinates of the points where C cuts or meets the coordinate axes.

(3)

(b) Write $f(x)$ in the form $ax^3 + bx^2 + cx + d$, where a, b, c and d are constants to be found.

(3)

(c) Hence, find the equation of the tangent to C at the point where $x = \frac{1}{3}$

(4)

(Total 10 marks)

Worked Solution - Question 6

1. Use the factorised form for the sketch

$f(x) = 2(x + 1)(x - 3)^2$. The curve cuts the x-axis at $(-1, 0)$ and meets the x-axis at $(3, 0)$ because $x = 3$ is a repeated root.

2. Find the y-intercept

$f(0) = 2(1)(9) = 18$, so the y-intercept is $(0, 18)$.

3. Describe the cubic shape

The leading term is positive, so the cubic goes down to the left and up to the right. It crosses at $x = -1$ and touches at $x = 3$.

4. Expand $f(x)$

$(x + 1)(x - 3)^2 = (x + 1)(x^2 - 6x + 9) = x^3 - 5x^2 + 3x + 9$, so
 $f(x) = 2x^3 - 10x^2 + 6x + 18$.

5. Differentiate

$f'(x) = 6x^2 - 20x + 6$.

6. Find the gradient at $x = 1/3$

$f'\left(\frac{1}{3}\right) = 6\left(\frac{1}{9}\right) - 20\left(\frac{1}{3}\right) + 6 = 0$, so the tangent is horizontal.

7. Find the y-coordinate

$f\left(\frac{1}{3}\right) = 2\left(\frac{4}{3}\right)\left(-\frac{8}{3}\right)^2 = \frac{512}{27}$.

8. Write the tangent

A horizontal tangent through this point has equation $y = \frac{512}{27}$.

Final answer

(a) x -intercepts $(-1, 0)$, $(3, 0)$, y -intercept $(0, 18)$.

(b) $f(x) = 2x^3 - 10x^2 + 6x + 18$.

(c) $y = \frac{512}{27}$.

Question 7

Radians

7.

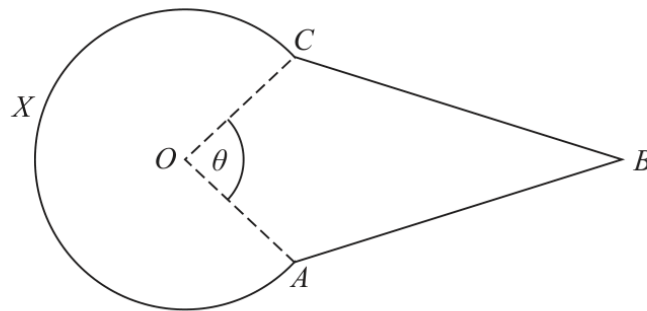


Figure 3

Figure 3 shows the design for a sign at a bird sanctuary.

The design consists of a kite $OABC$ joined to a sector $OCXA$ of a circle centre O .

In the design

- $OA = OC = 0.6$ m
- $AB = CB = 1.4$ m
- Angle $OAB = \text{Angle } OCB = 2$ radians
- Angle $AOC = \theta$ radians, as shown in Figure 3

Making your method clear,

- show that $\theta = 1.64$ radians to 3 significant figures, (4)
- find the perimeter of the sign, in metres to 2 significant figures, (2)
- find the area of the sign, in m^2 to 2 significant figures. (4)

(Total 10 marks)

Worked Solution - Question 7

1. Work in triangle OAB

In triangle OAB , $OA = 0.6$, $AB = 1.4$, and $\angle OAB = 2$ radians.

2. Find OB

By the cosine rule, $OB^2 = 0.6^2 + 1.4^2 - 2(0.6)(1.4) \cos 2$, so
 $OB = 1.73756 \dots$

3. Find half of theta

Using the cosine rule again, $\cos \angle AOB = \frac{0.6^2 + OB^2 - 1.4^2}{2(0.6)(OB)}$, giving
 $\angle AOB = 0.822199 \dots$ radians.

4. Use symmetry of the kite

The kite is symmetric, so $\theta = 2\angle AOB = 1.644398 \dots$, hence $\theta = 1.64$ radians to 3 significant figures.

5. Find the outside arc length

The sector $OCXA$ uses the major angle $2\pi - \theta$. Its arc length is
 $0.6(2\pi - 1.644398 \dots) = 2.78327 \dots$

6. Find the perimeter

Perimeter = $AB + CB + \text{major arc} = 1.4 + 1.4 + 2.78327 \dots = 5.58327 \dots$,
 so the perimeter is **5.6** m.

7. Find the kite area

The kite consists of two congruent triangles like OAB , so area of kite
 $= 2 \left(\frac{1}{2} (0.6)(1.4) \sin 2 \right) = 0.76381 \dots$

8. Find the sector area and total area

Major sector area $= \frac{1}{2}(0.6)^2(2\pi - 1.644398\dots) = 0.83498\dots$. Total area $= 0.76381\dots + 0.83498\dots = 1.59879\dots$, so the area is **1.6 m²**.

Final answer

(a) $\theta = 1.64$ rad.

(b) 5.6 m.

(c) 1.6 m².

Question 8

Straight Line

8.

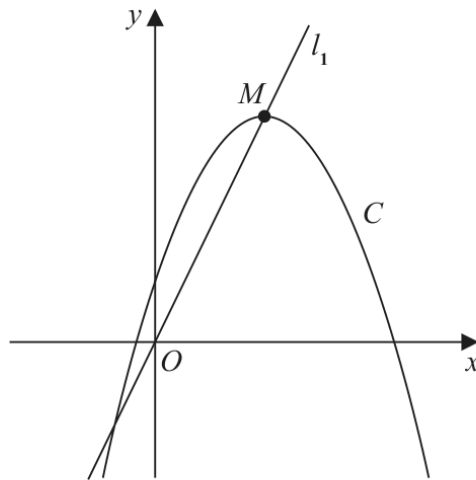


Figure 4

Figure 4 shows a sketch of the curve C with equation

$$y = 4 + 12x - 3x^2$$

The point M is the maximum turning point on C .

- (a) (i) Write $4 + 12x - 3x^2$ in the form

$$a + b(x + c)^2$$

where a , b and c are constants to be found.

- (ii) Hence, or otherwise, state the coordinates of M .

(5)

The line l_1 passes through O and M , as shown in Figure 4.

A line l_2 touches C and is parallel to l_1

- (b) Find an equation for l_2

(5)

(Total 10 marks)

Worked Solution - Question 8

1. Complete the square

$$4 + 12x - 3x^2 = -3(x^2 - 4x) + 4 = -3((x - 2)^2 - 4) + 4 = 16 - 3(x - 2)^2$$

2. State the maximum point

The maximum occurs when $(x - 2)^2 = 0$, so $M = (2, 16)$.

3. Find the gradient of l_1

Line l_1 passes through $O(0, 0)$ and $M(2, 16)$, so its gradient is $\frac{16}{2} = 8$.

4. Use the parallel tangent gradient

Line l_2 is parallel to l_1 , so its gradient is also 8. For $y = 4 + 12x - 3x^2$,
 $\frac{dy}{dx} = 12 - 6x$.

5. Find the point of contact

Set $12 - 6x = 8$, giving $x = \frac{2}{3}$. Then $y = 4 + 12\left(\frac{2}{3}\right) - 3\left(\frac{4}{9}\right) = \frac{32}{3}$.

6. Write l_2

Using gradient 8 through $\left(\frac{2}{3}, \frac{32}{3}\right)$, $y - \frac{32}{3} = 8\left(x - \frac{2}{3}\right)$. Hence
 $y = 8x + \frac{16}{3}$.

Final answer

$$(a)(i) \ 16 - 3(x - 2)^2. \quad (a)(ii) \ M = (2, 16).$$

$$(b) \ y = 8x + \frac{16}{3}.$$

WMA11/01 OCTOBER 2021

9 marks

Question 9

Transformations

9. In this question you must show all stages of your working.
Solutions relying on calculator technology are not acceptable.

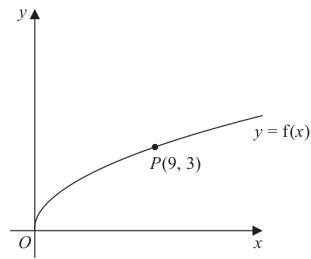


Figure 5

Figure 5 shows a sketch of the curve with equation $y = f(x)$ where

$$f(x) = \sqrt{x} \quad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

On the next page there is a copy of Figure 5 called Diagram 1.

- (a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q .

- (b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3(\sqrt{2} + 1)$$

(3)

- (c) Hence find, in simplest form, the coordinates of Q .

(3)

Question 9 continues

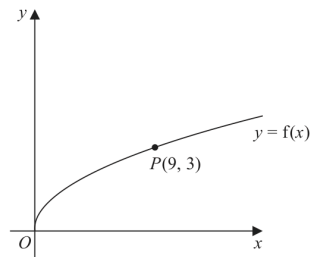
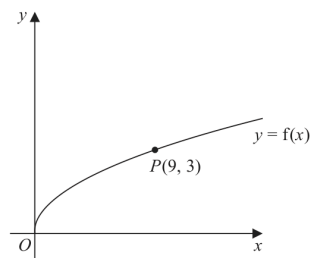


Diagram 1

Turn over for a copy of Diagram 1 if you need to redraw your graphs.

Only use this copy if you need to redraw your graphs.



Copy of Diagram 1

(Total 9 marks)

Worked Solution - Question 9

1. Transform P on $y = f(2x)$

$y = f(2x)$ is a horizontal scale factor of $\frac{1}{2}$, so $P(9, 3)$ maps to $\left(\frac{9}{2}, 3\right)$.

2. Transform P on $y = f(x) + 3$

$y = f(x) + 3$ is a translation 3 units upward, so $P(9, 3)$ maps to $(9, 6)$.

3. Set up the intersection equation

At Q , $f(2x) = f(x) + 3$. Since $f(x) = \sqrt{x}$, this gives $\sqrt{2x} = \sqrt{x} + 3$.

4. Show the required equation

Let $s = \sqrt{x}$. Then $\sqrt{2x} = \sqrt{2}s$, so $\sqrt{2}s = s + 3$. Hence $(\sqrt{2} - 1)s = 3$ and $s = \frac{3}{\sqrt{2} - 1} = 3(\sqrt{2} + 1)$.

5. Find the x-coordinate

$\sqrt{x} = 3(\sqrt{2} + 1)$, so $x = 9(\sqrt{2} + 1)^2 = 9(3 + 2\sqrt{2}) = 27 + 18\sqrt{2}$.

6. Find the y-coordinate

$y = \sqrt{x} + 3 = 3(\sqrt{2} + 1) + 3 = 6 + 3\sqrt{2}$. Therefore $Q = (27 + 18\sqrt{2}, 6 + 3\sqrt{2})$.

Final answer

(a) $P \mapsto \left(\frac{9}{2}, 3\right)$ on $y = f(2x)$, $P \mapsto (9, 6)$ on $y = f(x) + 3$.

(c) $Q = (27 + 18\sqrt{2}, 6 + 3\sqrt{2})$.

Question 10

Integration

10. A curve has equation $y = f(x)$, $x > 0$

Given that

- $f'(x) = ax - 12x^{\frac{1}{3}}$, where a is a constant
- $f''(x) = 0$ when $x = 27$
- the curve passes through the point $(1, -8)$

(a) find the value of a .

(3)

(b) Hence find $f(x)$.

(4)

(Total 7 marks)

Worked Solution - Question 10

1. Differentiate $f'(x)$

$$f'(x) = ax - 12x^{1/3}, \text{ so } f''(x) = a - 4x^{-2/3}.$$

2. Use $f''(27) = 0$

$$\text{Since } 27^{2/3} = 9, 0 = a - \frac{4}{9}, \text{ so } a = \frac{4}{9}.$$

3. Integrate $f'(x)$

$$\text{With } a = \frac{4}{9}, f'(x) = \frac{4}{9}x - 12x^{1/3}. \text{ Therefore } f(x) = \frac{2}{9}x^2 - 9x^{4/3} + c.$$

4. Use the point (1, -8)

$$-8 = \frac{2}{9} - 9 + c, \text{ so } c = -8 + 9 - \frac{2}{9} = \frac{7}{9}.$$

5. Write $f(x)$

$$f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}.$$

Final answer

$$(a) a = \frac{4}{9}.$$

$$(b) f(x) = \frac{2}{9}x^2 - 9x^{4/3} + \frac{7}{9}.$$